

From 230V AC to Adjustable 5V DC: Simulating and Designing a Linear Power Supply Using LM317A

EXPLORE MORE PROJECTS



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First Simulated & then Implemented physically

Performed by -

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As Stage – II of Power Supply Project

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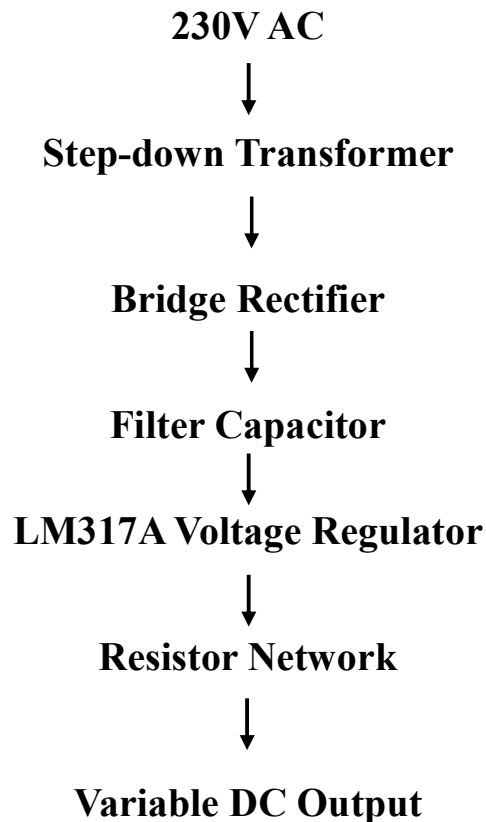
Why I Built This

After completing the **5V regulated supply using LM7805**, the next thing I wanted to try was **an adjustable regulated supply**.

The main idea was actually quite simple. In the previous circuit, the regulator was giving me a **fixed 5V** output. With **LM317A**, I wanted to understand how the output voltage can be changed using external resistors.

Circuit Arrangements

I kept most of the circuit same as the previous power supply -



The main change was the regulator section.

For LM317A, I used -

$$R_1 = 220\Omega$$

$$R_2 = 22\Omega$$

$$R_3 = 720\Omega \text{ (with the help of 10K preset in physical build)}$$

One thing I deliberately kept the same in simulation and hardware was the **220Ω + 22Ω resistance**. I didn't have the exact arrangement I wanted physically, so I used the same resistor values in simulation too. This made the comparison easier.

Understanding the LM317A

This was the part I actually wanted to understand rather than just connect according to a circuit diagram.

LM317A doesn't simply have a fixed output like **LM7805**. It maintains about **1.25V between OUT and ADJ**, and the external resistor network decides how much the output voltage increases.

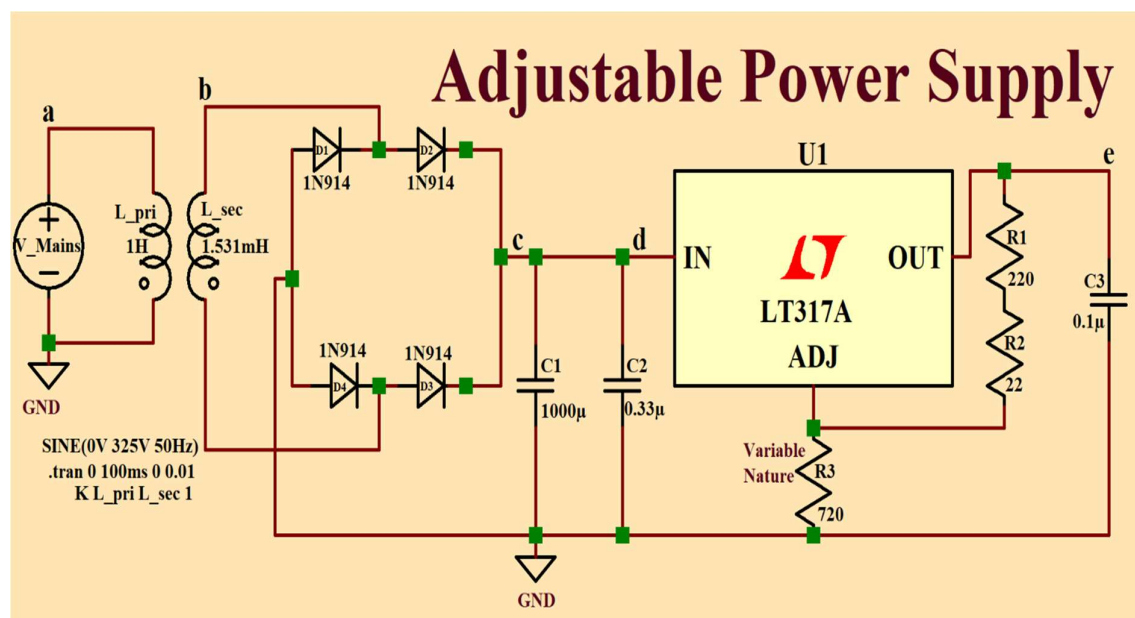
The basic equation is –

$$V_{out} = 1.25V \times \left(1 + \frac{R_2}{R_1}\right) + I_{Adj} \times R_2$$

So basically, **changing R₂ changes the output voltage**.

That is why the preset is connected in the adjustment network. By changing its resistance, one can change the output voltage.

Simulation of the Circuit



I first made the circuit in **LTspice** and checked the output.

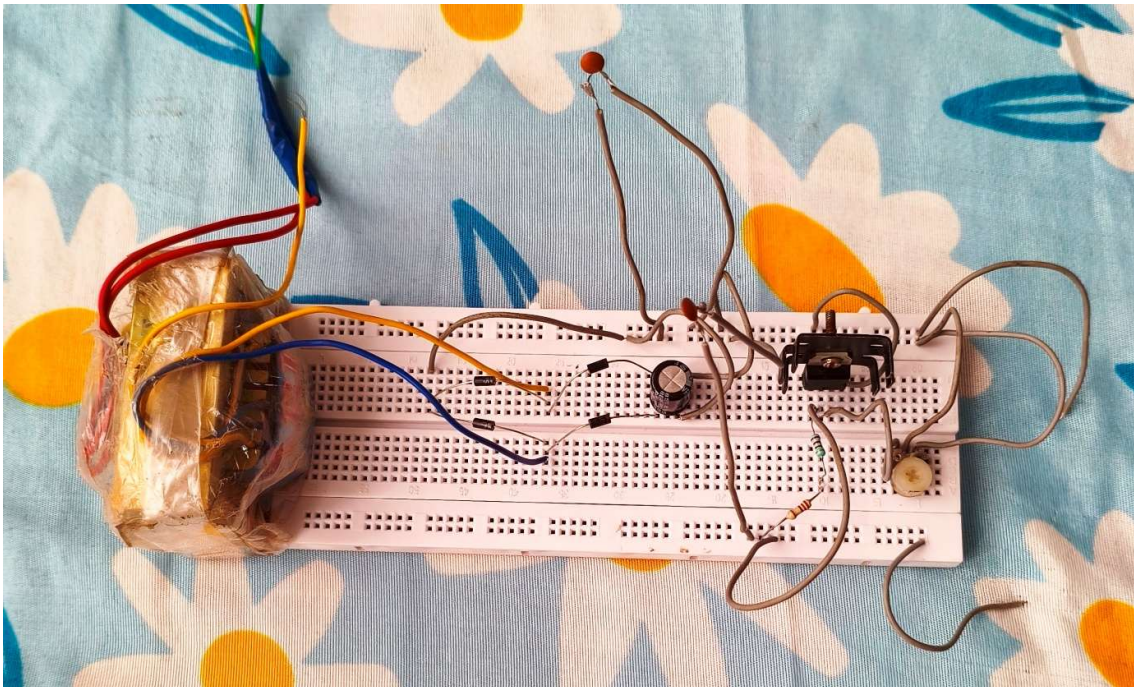
The transformer stepped the voltage down, the bridge rectifier converted AC into pulsating DC, and the 1000 μ F capacitor smoothed it before giving it to the LM317A.

After the regulator, I got a much smoother DC output, and by changing the resistance in the adjustment section, the output could be changed.

So at this point the concept was clear -

The transformer and rectifier provide the DC, while the LM317A and resistor network decides the regulated output voltage.

Building it physically



Then came the actual breadboard version.

I followed basically the same arrangement as my previous power supply project, so this part was much easier than starting from zero. The transformer, rectifier and filter section were already familiar to me.

The real headache this time was **adjusting the 10K preset to get 720 Ω .**

It was **extremely small**, and even when I measured it directly, it wasn't giving me exactly 720 Ω . While adjusting the circuit, the value was also fluctuating.

Eventually, one side of the preset seemed to have been partially burned, so it wasn't behaving properly anymore.

Because of that, the output wasn't as clean or stable as the simulation.

Still, I managed to get an output of around

$$V_{OUT} \approx 5.02V$$

Considering the condition of the preset and series connection of both resistor which was 239Ω , I was fully satisfied with the result.

I'm not going to spend another hour trying to make a faulty component behave like a laboratory instrument. I had already understood what I wanted from this stage.

Simulation vs Practical Results

Parameter	Simulation	Practical
Regulator	LT317A	LM317A
R1	220Ω	218Ω
R2	22Ω	23.4Ω
R3	720Ω Preset	$\approx 720\Omega$ Preset
Output	Stable/adjustable	$\approx 5.02V$
Main issue	All Ok	Getting 720Ω through 10K Preset
Additional	None	Heat Sink to secure the IC

The difference here was actually useful.

In LTspice, the preset is an ideal component. In the real circuit, my preset had **tolerance, mechanical variation and eventually some damage**. So the practical circuit obviously wasn't going to behave exactly like the simulation.

And honestly, this is one of the things I wanted to learn from these projects.

“Simulation tells me whether my circuit idea is correct, hardware tells me what happens when real components enter the picture.”

What I understood from this stage

The previous stage helped me understand the complete process of converting AC into a fixed regulated DC supply.

This stage added one more layer -

The output of a regulator doesn't always have to be fixed. It can be controlled using a feedback or resistor network.

I also got a much better feel for the **LM317A equation** because I wasn't just solving it on paper. I could actually change the resistance and see the effect on the output.

And the faulty preset, although annoying, was also part of the learning. It reminded me that a circuit working perfectly in simulation doesn't mean the physical circuit will automatically behave perfectly.

Challenges Faced

The most troublesome part of this experiment was the **10K Ω preset**.

The preset I had was very small and its resistance was not particularly stable. Even its direct resistance measurement was not exactly 720 Ω .

While adjusting the circuit, the output voltage fluctuated and eventually the preset showed **signs of burning**, with one side no longer behaving properly.

Instead of treating this as a complete failure, I considered it an important part of the practical experiment.

It showed me something that simulation cannot properly demonstrate -

A component having the correct nominal value does not guarantee that a physical circuit will behave perfectly.

Component condition, tolerance, mechanical adjustment, breadboard connections and even the way the component is handled can affect the result.

Conclusion

This was my first practical experiment with an adjustable linear voltage regulator.

By building the LM317A circuit, I understood how an external resistor network can control the regulator output and experienced the difference between an ideal simulated circuit and a real breadboard implementation.

The result was not perfectly clean because of the faulty adjustment preset, but I still achieved its main purpose. In fact, the imperfect result made the experiment more realistic and taught me something that a perfectly working simulation could not.

So Stage 2 is done.

Stage 1 - Fixed 5V regulated supply using LM7805

Stage 2 - Adjustable regulated supply using LM317A

And now the next question is -

How do I make the same voltage conversion much more efficient than a linear regulator?

That naturally leads toward switching regulators, buck converters and boost converters.